



Specialized Topics of Theoretical Physics

Majorana Modes in the Machine –
Simulating Topological Phases with Quantum Circuits
and AI

Measuring Topology on Circuits (VQE + String Order)

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1 Introduction

2 Variational preparation and circuit measurement

2.1 Hamiltonian, ansatz, and parity constraint

$$H(\mu) = -\frac{\mu}{2} \sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2} \sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2} \sum_{j=0}^{L-2} Y_j Y_{j+1}. \quad (2.1)$$

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0 \cdots 0\rangle, \quad \boldsymbol{\theta}^*(\mu) = \arg \min_{\boldsymbol{\theta}} C_{\mu}(\boldsymbol{\theta}), \quad (2.2)$$

$$P = \prod_{j=0}^{L-1} Z_j. \quad (2.3)$$

$$C_{\mu}(\boldsymbol{\theta}) = \langle H(\mu) \rangle_{\boldsymbol{\theta}} + \lambda (1 - \langle P \rangle_{\boldsymbol{\theta}})^2, \quad \lambda = 3, \quad (2.4)$$

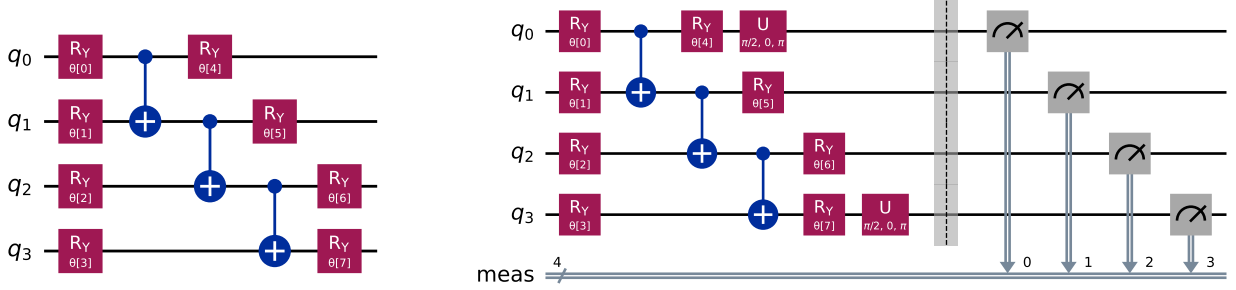


Figure 1:

2.2 Why the observable must be non-local

$$\langle X_0 \rangle = \langle \psi | P X_0 P | \psi \rangle = -\langle X_0 \rangle = 0. \quad (2.5)$$

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1} = i b_0 a_{L-1}, \quad [O_{\text{edge}}, P] = 0, \quad (2.6)$$

2.3 Basis rotation and shot estimator

$$m_s = (-1)^{\sum_{j=0}^{L-1} b_j^{(s)}}, \quad \widehat{\langle O_{\text{edge}} \rangle} = \frac{1}{N} \sum_{s=1}^N m_s. \quad (2.7)$$

3 Methods and results

3.1 Numerical protocol and validation

quantity	value
chain and couplings	$L = 4, t = \Delta = 1$
sweep	81 points, $\mu/t \in [-3.5, 3.5]$
ansatz	R_Y + linear CNOT, $r = 3, N_\theta = 16$
optimizer	COBYLA, at most 500 iterations per start
symmetry penalty	$\lambda(1 - \langle P \rangle)^2$, $\lambda = 3$
continuation	previous optimum as the next initial point
recovery	up to 4 additional random restarts
readout	4096 shots for O_{edge}
acceptance	$ \Delta E \leq 0.05, 1 - \langle P \rangle \leq 0.01, \mathcal{F} \geq 0.99$

Table 1:

3.2 Local versus non-local readout

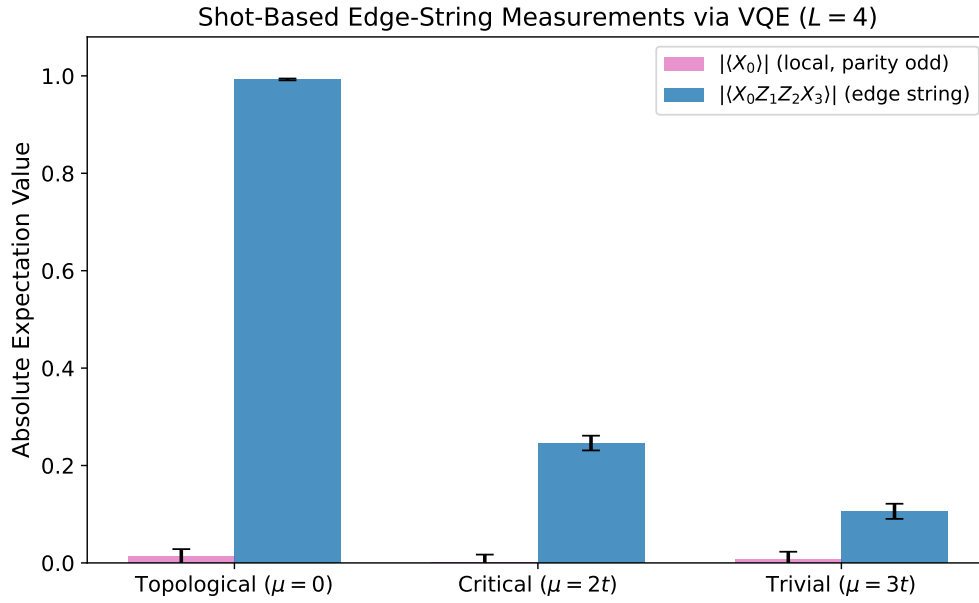


Figure 2:

3.3 Parity-constrained VQE across the phase sweep

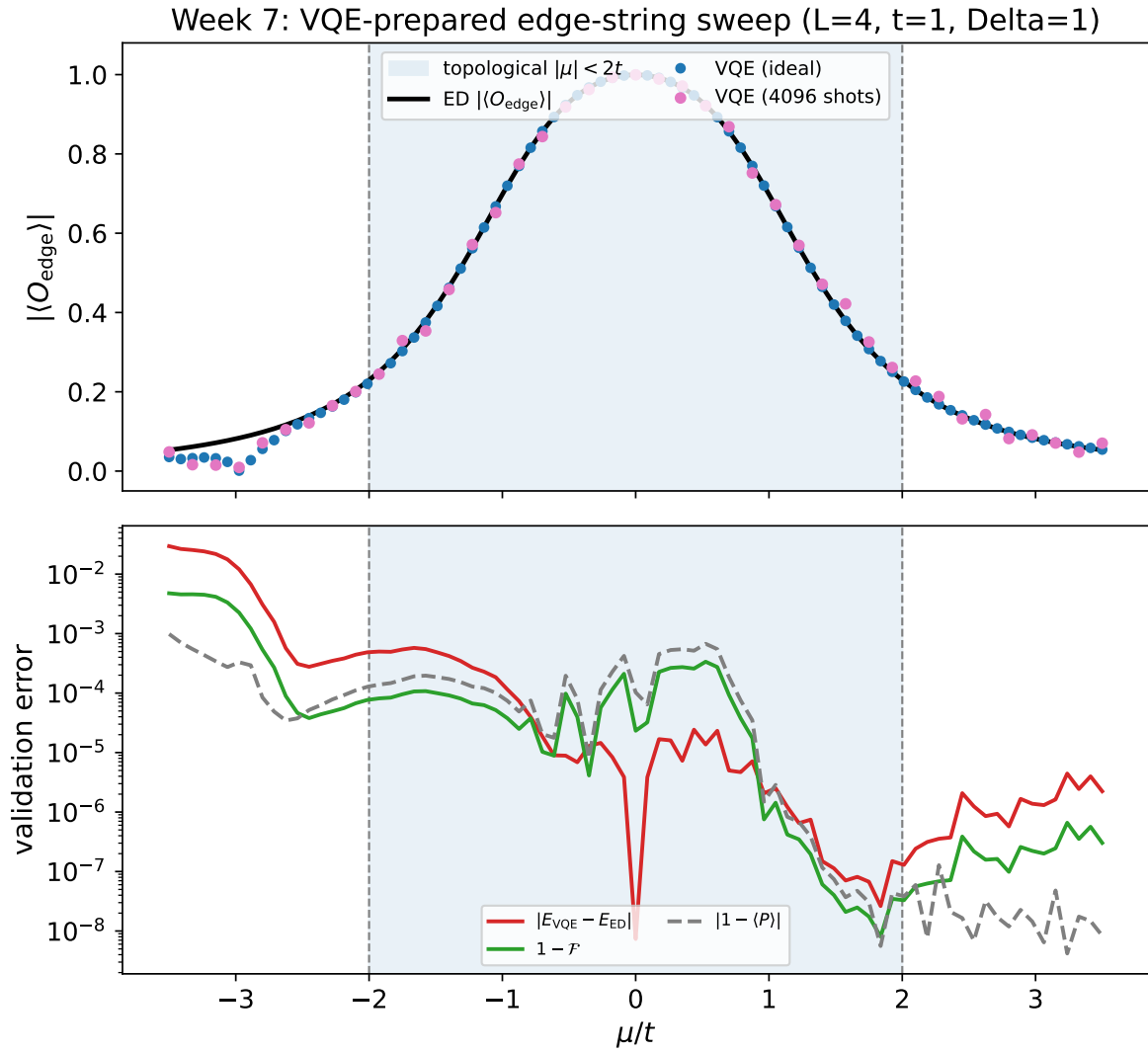


Figure 3:

3.4 How deep must the noiseless ansatz be?

Week 7: ansatz-depth diagnostic ($L = 4$, $t = \Delta = 1$)

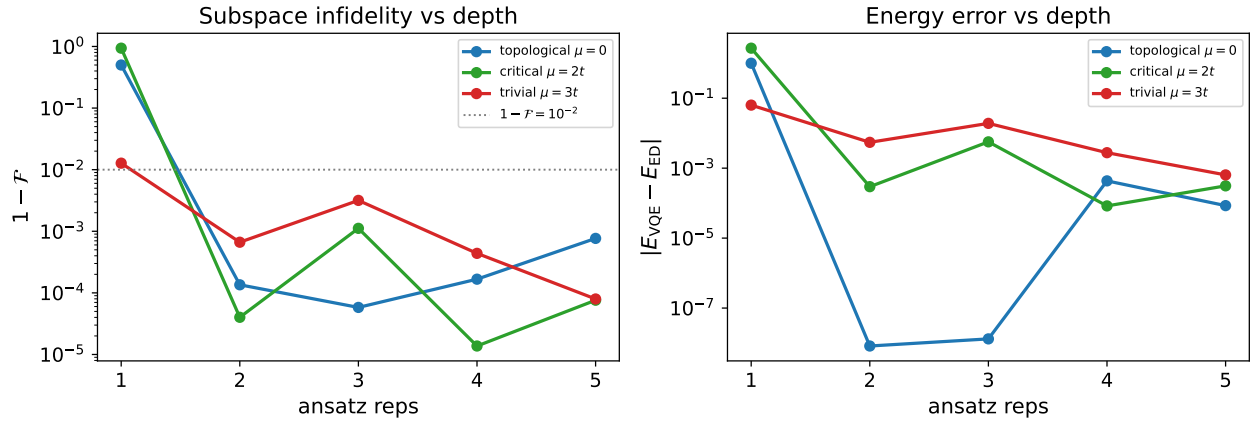


Figure 4:

4 On the AI collaboration